2 Part 2 – Converters

2.1 Assessment Requirements – Converters

You are an electrical engineer working as part of a power electronics and drives consultancy. A customer has commissioned a project aimed at identifying a suitable DC-link capacitor for a three-phase motor drive they are designing for a large industrial fan. They have down-selected two candidate capacitors, shown in Table 2-1, which you are required to evaluate and subsequently recommend the component that is most appropriate for their system. Table 2-2 shows a list of operating parameters that have been provided to you by the company for their system.

Provide a brief report that includes:

- For the selected component, four annotated waveforms showing: *(ILO1 and ILO2)*
  1. DC-link inductor current Vs time profile
  2. DC-link inductor voltage Vs time profile
  3. Capacitor current Vs time profile
  4. Capacitor Voltage Vs time profile
- A statement on how you derived confidence in the final result *(ILO3)* – up to 100 words
- A judgement on the design task *(ILO4)* – up to 50 words

Table 2-1: Down-selected capacitor options

Capacitor option 1:		Capacitor option 2:	
Capacitance		Capacitance	
ESR		ESR	
Ripple current rating		Ripple current rating	

Table 2-2: Drive operating parameters

Parameter	Value
Vd	-44.7
Vq	105.2
Iphpk	110A (Id=0A)
Switching frequency	10000Hz
Modulation Scheme	SPWM
Maximum Vdc ripple	3%
Supply voltage	325V
Fundamental Frequency (f1)	40Hz

2.2 Example solution - Converters:

2.2.1 Key attributes (Things we are looking for when determining a mark)

1. L_{dc} current waveform with an appropriate annotation
2. L_{dc} voltage with an appropriate annotation
3. Capacitor current Waveform – Annotation shows I_{ripple}
4. Capacitor voltage waveform – Annotation shows V_{DC} ripple ($V_{esr} + V_c$)
5. Correct statement outlining the suitability of the two DC-link capacitor options (50 words).
6. Statement (around 100-200 words) clearly communicating how you have derived confidence in your final results and elaborates any results you are less confident in and why. Try and write this as if you were trying to convince a chief engineer that need to approve or sign off your results.

2.2.2 Demonstration – Converters

2.2.2.1 Analysis/Simulation Outcomes (ILOs 1 & 2)

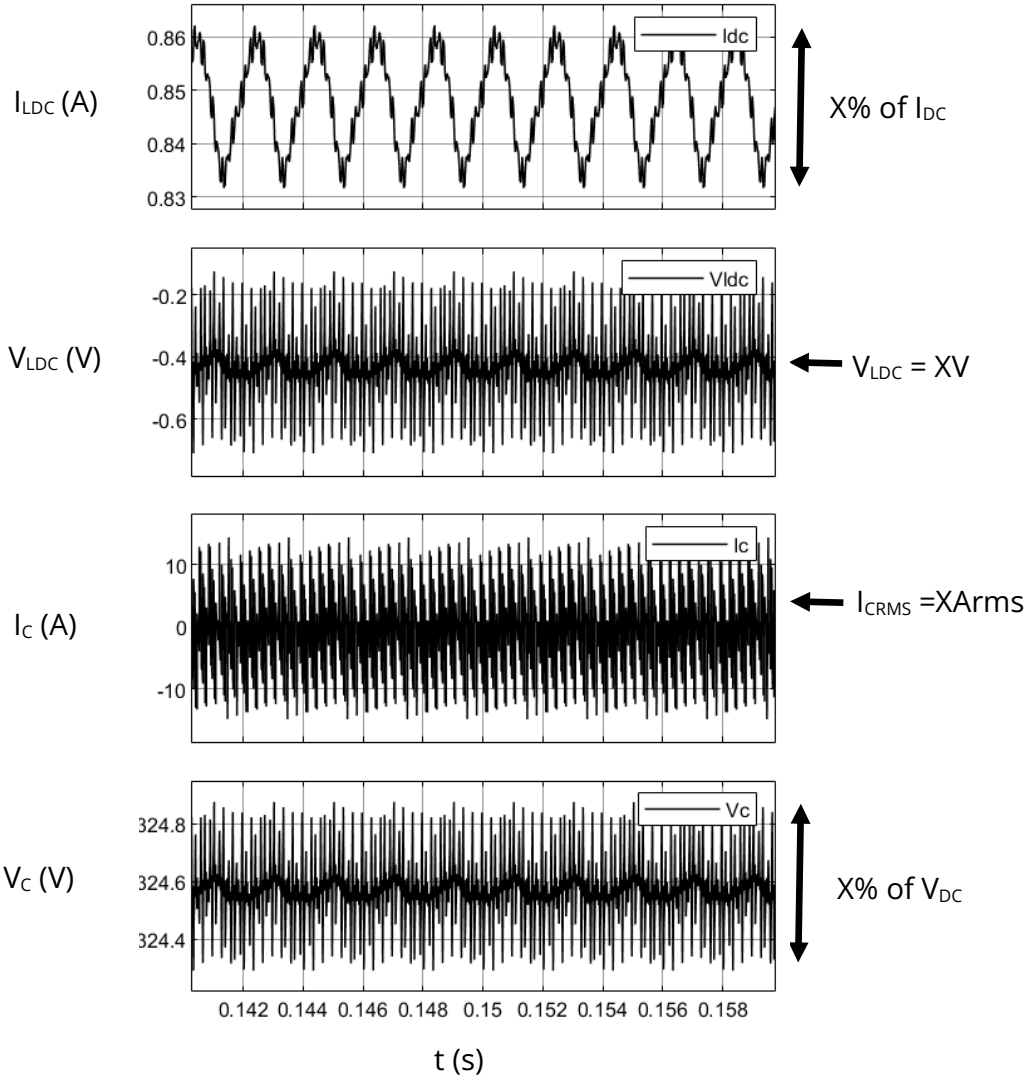


Figure 2-1: Example Figure Submission for Part 2 (not the actual answer)

2.2.2.2 DC Link Suitability (ILO 3 – Pass/Fail)

Capacitor option X parameters meets the requirements whereas Capacitor option Y does not due to….

2.2.2.3 Communicate How You Have Derived Confidence in your Results (ILO 4 – Pass/Fail)

I derived confidence in my results using the follow technique(s)/approach(es) … .

I'm less certain in the following results because….